MILITARY SPECIFICATION

ADAPTERS, CONDUCTIVE NEOPRENE, ENDOTRACHEAL ANESTHESIA SET

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers conductive adapters suitable for use with standard endotracheal anesthesia circuits.

1.2 Classification. Adapters shall be of the following types and styles as indicated below (see 6.2):

Types	Styles
I - Right Angle Elbow	1 - Without Pop-Off Valve
II - Catheter "Y"	2 - With Pop-Off Valve
III - Chimney "Y"	

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant (for Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24228 ADAPTERS, ENDOTRACHEAL ANESTHESIA SET

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I. Z79.2 Endotracheal-Tube Connectors and Adapters: Specifications For Anesthetic Equipment.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N. Y. 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 56 Code for Use of Flammable Anesthetics (Safe Practice For Hospital Operating Rooms.)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Stret, Boston, Massachusetts 02110.)

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3. REQUIREMENTS

3.1 Fire and casualty hazards. The adapters specified herein shall comply with the fire and casualty hazards provisions of the applicable standards for operation in hazardous locations, class 1, Group C, and as prescribed by the National Fire Protection Association NFPA No. 56, and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp, or label to the article, or issuance of a certificate. The stamp, label, seal or certificate of the Underwriter's Laboratories, Inc., or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of such approval. However, in addition to the aforementioned, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance.

3.1.2 Compliance with the above requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material.

3.2.1 Synthetic rubber compounds. Adapters shall be fabricated from conductive neoprene capable of withstanding repeated steam sterilization without deformation. Material shall not permanently deform, or be adversely acted upon, with medicinal gases generally used in anesthesia procedures.

3.2.2 Other material. Inserts of Type II adapters shall be fabricated from unplated martensitic or austenitic corrosion-resisting steel. Style 2 adapters shall have valve components fabricated from corrosion-resisting materials suitable for the purpose intended. External components of the valves fabricated from brass shall be chromium plated.

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3.3 Style, design and dimensions. Adapters specified herein shall be conductive and suitable for use in standard anesthesia procedures involving medicinal gases. Adapters shall be of durable construction and fabricated from material specified in 3.2. All adapters shall be sterilizable by standard hospital procedures. Adapters shall have a 15 mm female internal taper, in accordance with figure 5 of A.N.S.I. Z79.2, at the patient end (portion closest to the patient in the anesthesia circuit), as shown on drawing. Adapters shall be in strict accordance with DPSC Drawing 24228 for appropriate type, style, stock number and item identification.

3.3.1 Type I adapter. Type I adapter shall be a right angle elbow adapter suitable for use with oronasal maks, and between 15 mm. endotracheal tube connectors and Y-piece adapters when used with standard anesthesia apparatus. Type I adapter shall not have pop-off valve.

3.3.2 Type II adapter. Type II adapter shall be a Y-piece, catheter "Y", suitable for use between FSN 6515-299-8710 Adapter, Right Angle Elbow, or Type I adapters specified above, or a 15 mm. endotracheal tube connector and FSN 6515-302-1500 Tube, Inhaler of a standard anesthesia apparatus. Type II adapters shall have an integral corrosion-resistant metal insert (not plated) at the 15 mm taper, patient end.

3.3.3 Type III adapter. Type III adapter shall be a Y-piece, chimney "Y", suitable for use in inhalation anesthesia, concerning: oronasal masks, 15 mm endotracheal connectors; and inhaler tubes (FSN 6515-302-1500).

3.3.4 Style 2 adapter. Style 2 adapters shall apply only to Type II and Type III adapters specified herein. Style 2 adapters shall be as specified in 3.3.2 and 3.3.3, except that the adapters shall contain "pop-off" valves, located in proximity of the patient end. Valves shall be fabricated from material specified in 3.2.2. Valves shall be of the screw design, capable of manual operation in order to bleed off excessive quantities of anesthesia gases to atmosphere. Valve components, in addition to the material proper (neoprene) of the adapters, shall be capable of withstanding repeated steam sterilization. Screw component of the adapter shall be contained by a suitable locking ring member.

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3.4 Electrical conductivity. Adapters shall be electrically conductive and possess electrical characteristics specified in NFPA No. 56, when tested in accordance with Article 254 of NFPA No. 56.

3.5 Sterilization. All adapters, including valve components of style 2, shall be capable of autoclaving without indication of damage, separation, or deformation when tested as specified in 4.4.1. In addition, all material shall not show evidence of corrosion as a result of this test.

3.6 Accelerated aging. Neoprene material of all adapters, specified herein, shall withstand the accelerated aging test specified in 4.4.2 without becoming tacky or brittle. Further, valve components of style 2 adapters, shall not separate from neoprene material as a result of this test.

3.7 Identification markings. Adapters shall be permanently and legibly marked with the name or registered trademark of the contractor.

3.8 Workmanship. Workmanship shall be first class throughout. Ends of adapters shall be finished smooth and free from excessive or interfering flash. External surfaces of valves (type 2 adapters) shall be finished bright and free from burrs and sharp edges. Adapters shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the neoprene, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one adapter.

Table I. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one adapter.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test Procedure	Inspection level	AQL
Adapter	Dimensions	3.3	- - -	S-1	*
	Electrical Conductivity test	3.4	NFPA No. 56	S-1	*
	Sterilization test	3.5	4.4.1	S-1	*
	Accelerated aging test	3.6	4.4.2	S-1	*

* Acceptance number shall be zero for all sample sizes.

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4.3 Examination. The adapters shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any adapters in the sample containing one or more defects shall be rejected and if the number of defective adapters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Adapters not of the type and style specified.
102	Adapters not free from burrs, slivers, sharp edges, pits, holes, or excessive flash.
103	Valve components of type 2 adapters not secured to body of adapter.
104	Outer surfaces of adapters not smooth or free from tool marks.
105	External surfaces of valve (type 2 adapters) not plated, or finish not bright.
106	Patient end of Type II adapter does not contain internal metal insert.
Minor	
201	Adapters not permanently and legibly identified as specified in 3.7.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterilization tests. Adapters shall be placed in a suitable sterilizer (container) and steam sterilized at a temperature of 250° Fahrenheit (°F) for a period of 10 minutes or 272° F for a period of 3 minutes. The adapters shall then be removed from the sterilizer and allowed to cool to room temperature. This test shall be performed a total of 10 times. The appearance of the adapters shall be noted.

4.4.2 Accelerated aging test. Adapters shall be subjected to a temperature of 158° ± 2° F for seven days in an air oven. The appearance of the adapters after completion of this test shall be noted.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One adapter, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling. As an alternate, each unit may be packaged in a polyethylene bag of appropriate size. Closure of bag shall be effected by heat sealing.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. These adapters of conductive neoprene are to be used in conjunction with standard 15 mm endotracheal connectors, and other components normally used in standard anesthesia gas dispensing circuits.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title and date of this specification.
(b) Type and style (see 1.2).
(c) Applicable Federal Stock Number (FSN)
(d) Selection of applicable level of packaging and packing required.

6.3 This specification covers the following items listed in the Federal Supply Catalog:

FSN	Item Identification
6515-880-1832	ADAPTER, RIGHT ANGLE ELBOW, ENDOTRACHEAL ANESTHESIA SET
6515-880-1833	ADAPTER, Y-PIECE, ENDOTRACHEAL ANESTHESIA SET, CATHETER "Y".
6515-880-1834	ADAPTER, Y-PIECE, ENDOTRACHEAL ANESTHESIA SET, CHIMNEY "Y".
6515-880-1835	ADAPTER, Y-PIECE, ENDOTRACHEAL ANESTHESIA SET, CATHETER "Y", with POP-OFF VALVE.
6515-880-1836	ADAPTER, Y-PIECE, ENDOTRACHEAL ANESTHESIA SET, CHIMNEY "Y" with POP-OFF VALVE.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1468

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-36665 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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